Proof of work · 3 production systems

Full stack case studies. Built, shipped, measured.

01 **Intimec IGA**
Enterprise identity governance platform · Next.js + Spring Boot microservices

02 **Cunocovo**
Social platform for creative professionals · Next.js + Node.js + PostgreSQL

03 **GovHub**
HR platform for a 15,000-employee municipality · Next.js + NestJS + Flutter

Muhammad Talha Atif();

Intimec IGA
Self-service access, audit-ready by design

THE PROBLEM

Access to systems ran on email and spreadsheets. Employees could not self-serve, bulk changes needed engineers, and audit trails were incomplete.

WHAT I BUILT

- The **Access Portal** end to end: tile catalog, cart, multi-user requests, approval routing and provisioning
- A generic **rule-based engine** that assigns roles and entitlements through grouped conditions, CRON and event scheduling
- Event-driven approvals on **Kafka** with dead-letter recovery, so no event is lost silently
- PII-free **audit logging** capturing before and after state for every transition

MEASURED RESULTS

-50%
inter-service latency after REST to gRPC migration

5,000+
users per bulk role operation

5M+
documents migrated with zero downtime

100K+
analytics logs per cycle after schema redesign

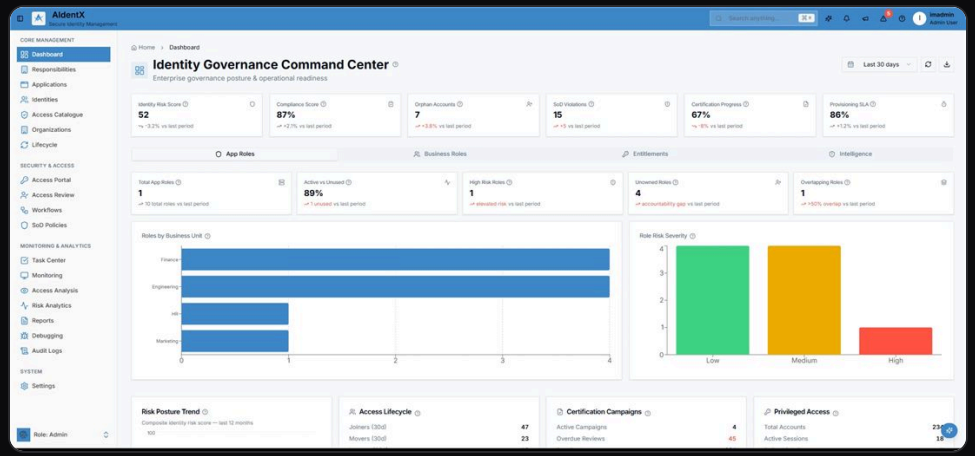
STACK

Next.js · TypeScript · Tailwind · TanStack Query · Spring Boot · Kafka · gRPC · PostgreSQL · OpenSearch · Docker · GitHub Actions

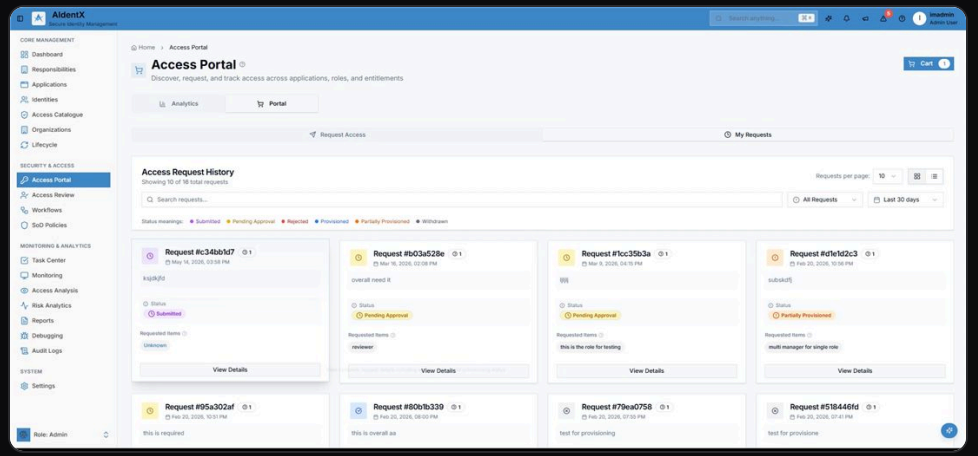
Role: owned the Access Portal; contributed across 8+ microservices

Live: aidentx.com/igax

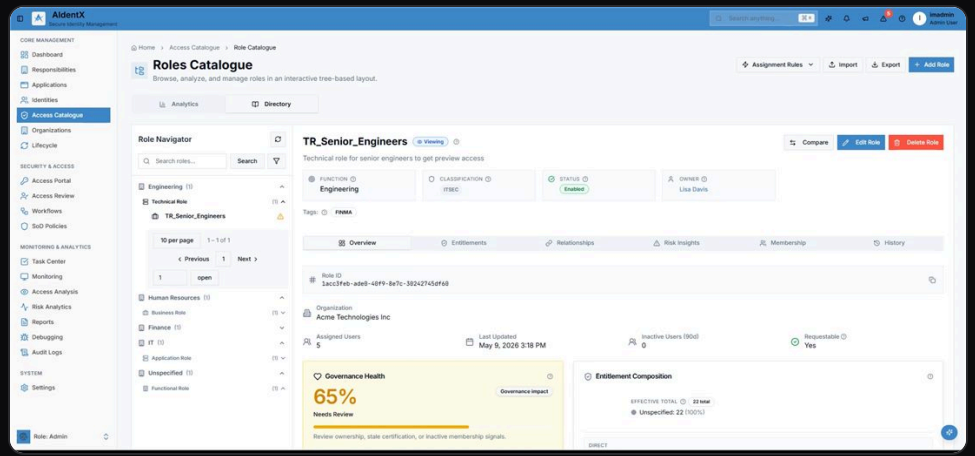
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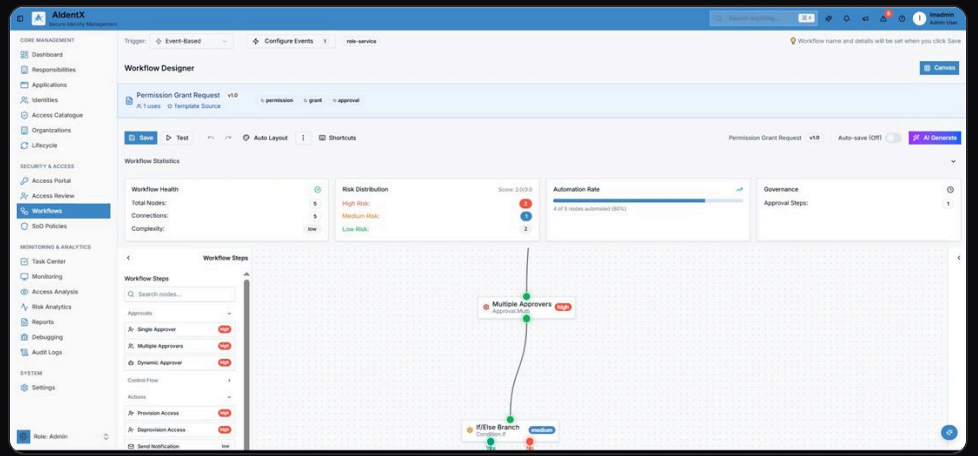
Identity Governance command center: risk and compliance at a glance



Access Portal: request history and status tracking



Role catalogue with hierarchy, health and membership



Workflow designer for approval chains

Screens from the Intimec IGA platform (test environment)

Next.js dashboard, standardised theme and component library

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Cunocovo
A feed that loads in 700ms, not 3 seconds

THE PROBLEM

A live social platform for photographers and videographers had a slow feed, duplicate notifications, choppy scrolling on iOS and Android, and a moderation process that needed an engineer for every report.

WHAT I FIXED AND BUILT

- Feed: found **four separate causes** (missing composite index, 9 unused columns, query shape, no cache) and fixed each layer
- Notification duplicates: replaced a race condition with a **transaction and unique constraint**
- Mobile and Safari lag: passive listeners, Intersection Observer, real `button` elements
- End-to-end **moderation system** and admin panel, plus IP rate limiting
- **AWS EC2 to Fly.io** migration with Docker and DNS cutover, no downtime

MEASURED RESULTS

3s → 700ms
feed API latency (77% faster)

2s → 0ms
comment button response

0
minutes of downtime during infrastructure migration

0
duplicate notifications after the transactional rewrite

STACK

Next.js · React · TypeScript · Node.js · Express · PostgreSQL · Drizzle ORM · Supabase · WebSockets · JWT + Google OAuth · Docker · Fly.io

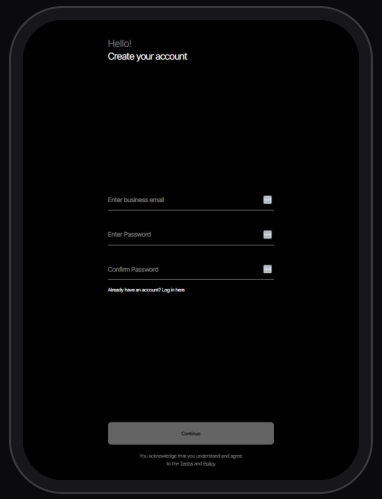
Role: full stack contributor on a live production platform

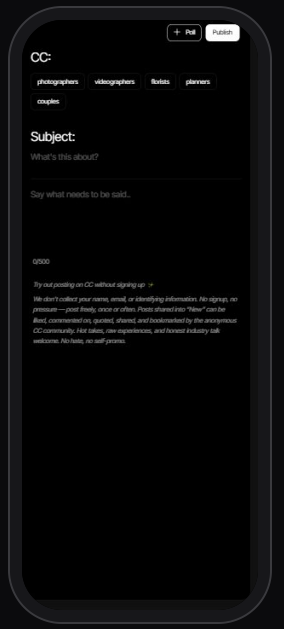
Live: uppercase.cc

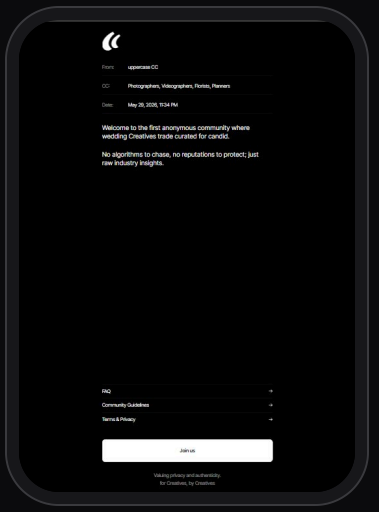
Muhammad Talha Atif();



Mobile-first sign-up, posting and community flows







Next.js client + separate admin panel, Express API, PostgreSQL via Drizzle

2,000+ registered users

Muhammad Talha Atif();

GovHub
HR and attendance for 15,000 municipal employees

THE PROBLEM

A South African municipality ran attendance, leave and study assistance on paper forms and spreadsheets. Approvals took weeks and compliance reporting could not be automated.

WHAT I BUILT

- NestJS backend in **12 domain modules** with zero circular dependencies
- A **configurable approval-chain engine**: HR changes routing from the admin panel, no deployment
- Next.js admin panel with **role-scoped** views and Flutter app with offline clock-in sync
- **Encrypted bulk import** (AES-256-GCM, BullMQ) with per-row failure isolation
- Append-only **audit log** and a CI pipeline gating every merge

SCOPE

15,000
employees the platform is built to serve

12
domain modules and 12 admin pages

9
leave types with entitlement rules

4
approval levels, configurable per role

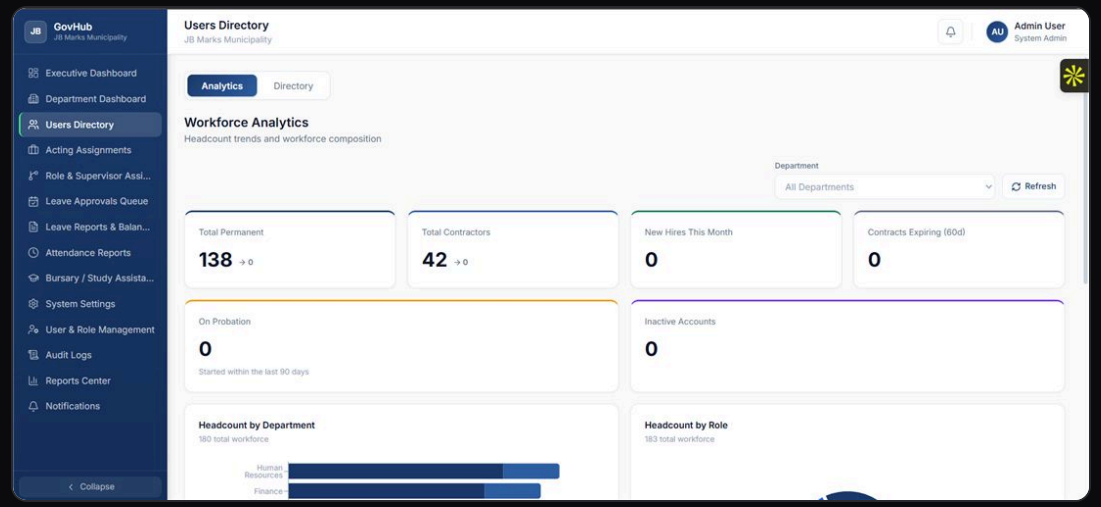
STACK

Next.js 14 · TypeScript · Flutter · NestJS · Prisma · PostgreSQL · Redis · BullMQ · Cloudflare R2 · Docker · GitHub Actions

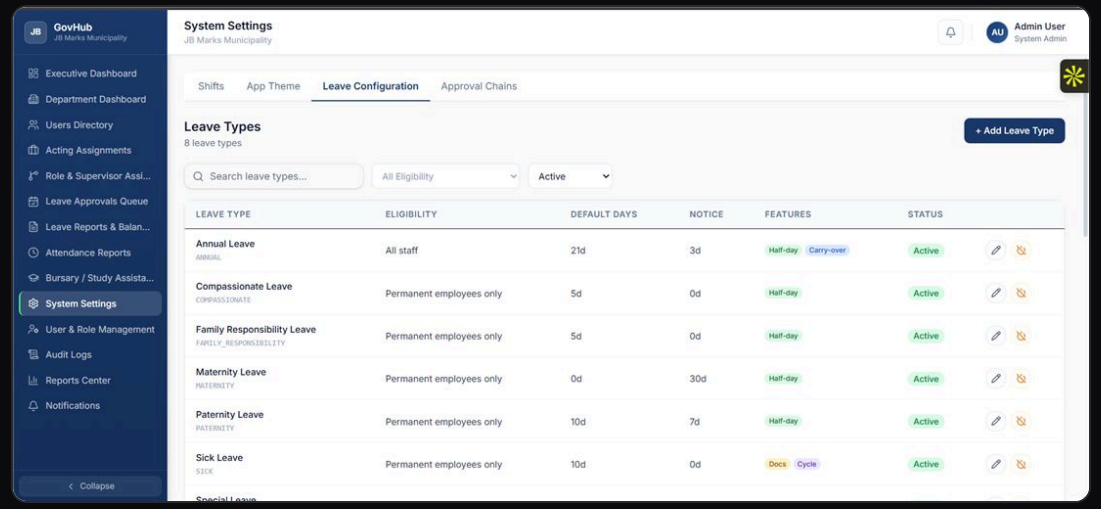
Role: Founding Engineer (Contract), API, admin panel, infrastructure

Live: mystaffhub.org

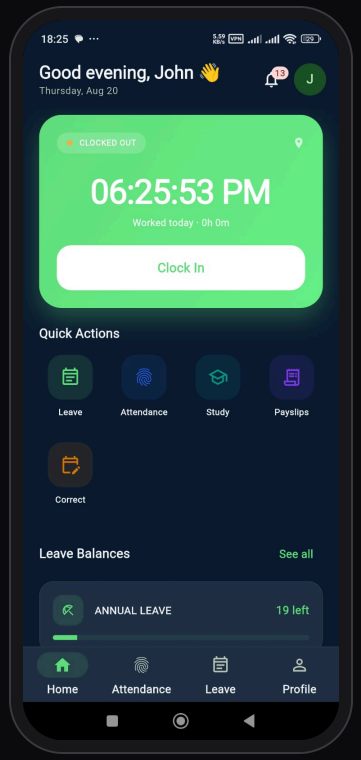
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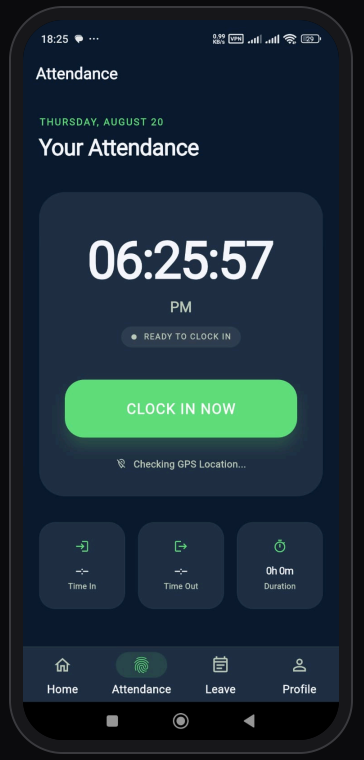


Workforce analytics, Redis-cached aggregations



Leave types configuration for HR admins







Next.js admin panel and Flutter mobile app (demo data)

Offline-first attendance, multi-level approvals